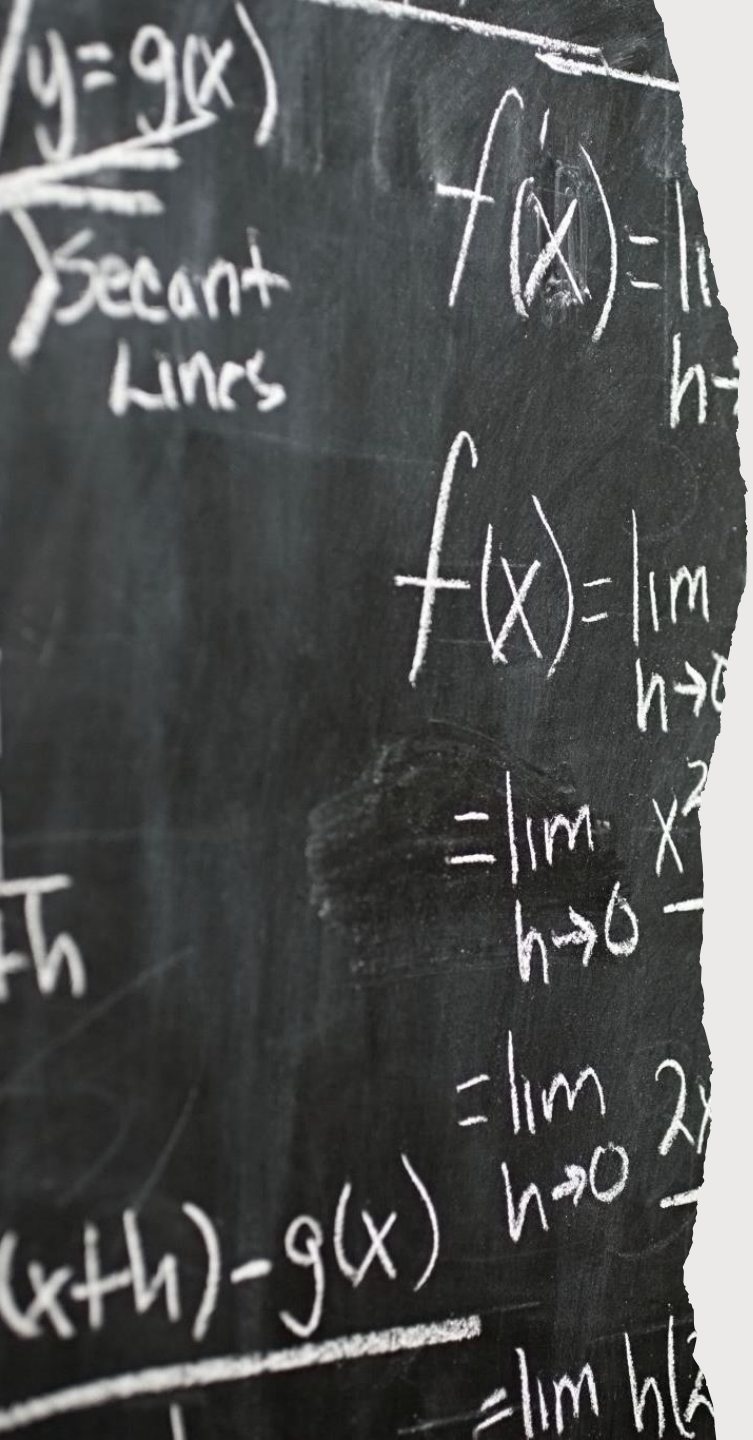


Mathematics 1

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Syllabus

I. Algebra:

- ❑ 2nd degree equations and systems
- ❑ Parametrical equations
- ❑ Radical equations and functions
- ❑ Logarithms equations and functions
- ❑ Series; concept, Fourier's Series, Taylor's Series

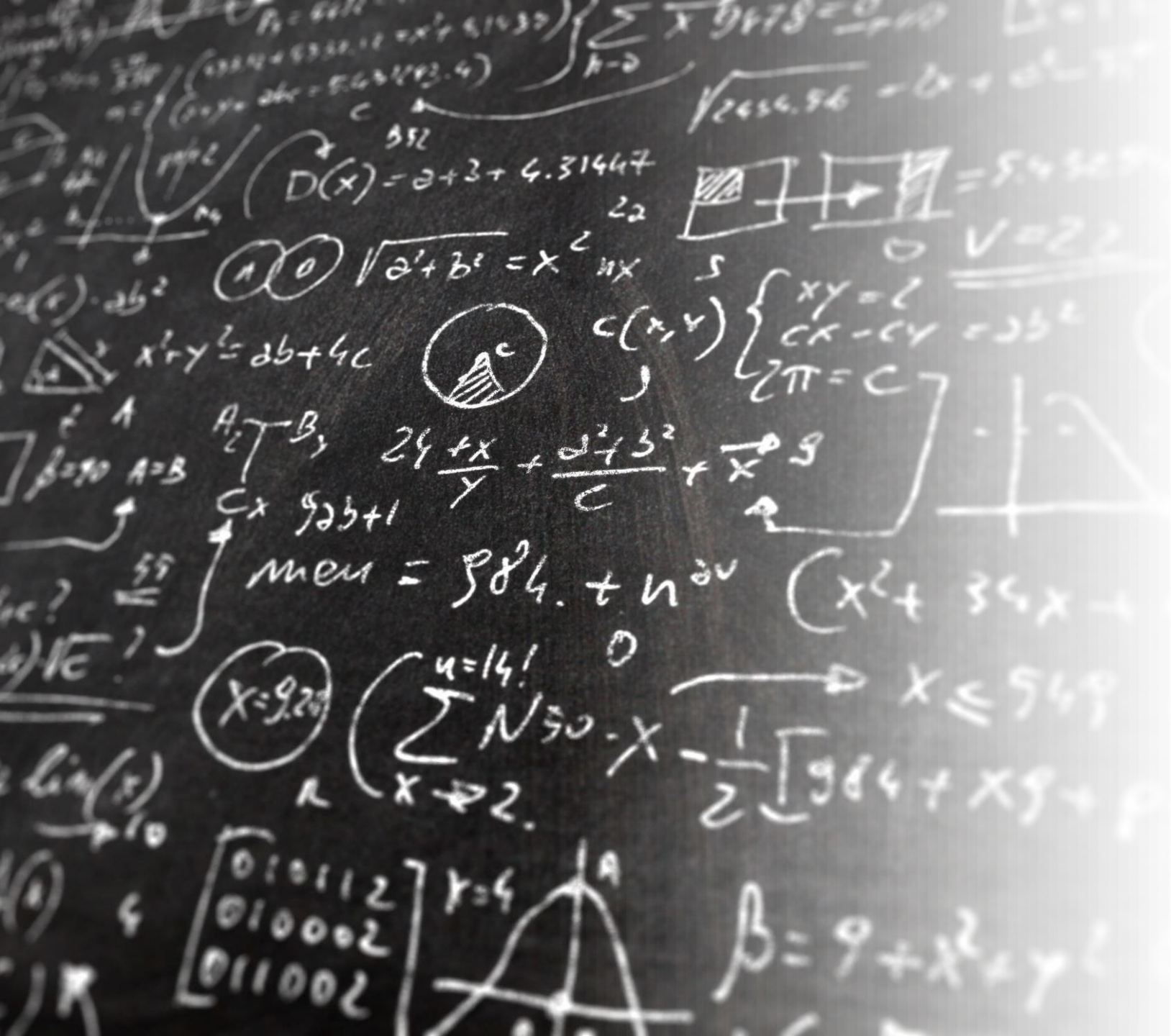
II. Trigonometry:

- ❑ Sin and Cosine theory
- ❑ Trigonometry equation & functions
- ❑ Applications of trigonometry

III. Analytical geometry:

- ❑ Distance between two points, a point and a segment
- ❑ 1st degree equation (line)
- ❑ 2nd degree equations (Parabola)
- ❑ Intersection, Parallelism and Tangent

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I. Algebra

Chapte1: Prerequisites

1. Real Numbers: Algebra Essentials

Sets of Numbers :

Natural numbers	$\mathbb{N} = \{1, 2, 3, 4, \ldots\}$.
Integers	$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
Rational numbers	$\mathbb{Q} = \left\{ \frac{p}{q} \;\middle|\; p, q \in \mathbb{Z};\ q \neq 0 \right\}$.
Irrational numbers	$\mathbb{Q}' = \{\ldots, \sqrt{2}, \sqrt[3]{-5}, \pi, e, \ldots\}$.
Real numbers	$\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'$.

Intervals :

Notation	Set description	Picture
(a,b)	$\{x \mid a < x < b\}$	
$[a,b]$	$\{x \mid a \le x \le b\}$	
$[a,b)$	$\{x \mid a \le x < b\}$	
(a,∞)	$\{x \mid x > a\}$	
$(-\infty,b]$	$\{x \mid x \le b\}$	
$(-\infty,\infty)$	$\mathbb{R} = \{x \mid -\infty < x < \infty\}$	

properties of real numbers

The following properties hold for real numbers a, b, and c.

	Addition	Multiplication
Commutative Property	$a + b = b + a$	$a \cdot b = b \cdot a$
Associative Property	$a + (b + c) = (a + b) + c$	$a(bc) = (ab)c$
Distributive Property	$a \cdot (b + c) = a \cdot b + a \cdot c$	
Identity Property	There exists a unique real number called the additive identity, 0, such that, for any real number a $a + 0 = a$	There exists a unique real number called the multiplicative identity, 1, such that, for any real number a $a \cdot 1 = a$
Inverse Property	Every real number a has an additive inverse, or opposite, denoted $-a$, such that $a + (-a) = 0$	Every nonzero real number a has a multiplicative inverse, or reciprocal, denoted $\frac{1}{a}$, such that $a \cdot \left(\frac{1}{a}\right) = 1$

the product rule of exponents

For any real number a and natural numbers m and n, the product rule of exponents states that

$$a^m \cdot a^n = a^{m+n}$$

For example:

- $t^5 \cdot t^3 = t^{5+3} = t^8$
- $(-3)^5 \cdot (-3) = (-3)^5 \cdot (-3)^1 = (-3)^{5+1} = (-3)^6$
- $x^2 \cdot x^5 \cdot x^3 = x^{2+5+3} = x^{10}$

the quotient rule of exponents

For any real number a and natural numbers m and n, such that $m > n$, the quotient rule of exponents states that

$$\frac{a^m}{a^n} = a^{m-n}$$

For example:

- $\dfrac{(-2)^{14}}{(-2)^9} = (-2)^{14-9} = (-2)^5$
- $\dfrac{t^{23}}{t^{15}} = t^{23-15} = t^8$

the power rule of exponents

For any real number a and positive integers m and n, the power rule of exponents states that

$$(a^m)^n = a^{m \cdot n}$$

For example:

- $(x^2)^7 = x^{2 \cdot 7} = x^{14}$
- $((2t)^5)^3 = (2t)^{5 \cdot 3} = (2t)^{15}$
- $((-3)^5)^{11} = (-3)^{5 \cdot 11} = (-3)^{55}$

the zero exponent rule of exponents

For any nonzero real number a, the zero exponent rule of exponents states that

$$a^0 = 1$$

For example:

- $\dfrac{c^3}{c^3} = c^{3-3} = c^0 = 1$

the negative rule of exponents

For any nonzero real number a and natural number n, the negative rule of exponents states that

$$a^{-n} = \frac{1}{a^n}$$

For example:

- $\frac{\theta^3}{\theta^{10}} = \theta^{3-10} = \theta^{-7} = \frac{1}{\theta^7}$
- $\frac{z^2 \cdot z}{z^4} = \frac{z^{2+1}}{z^4} = \frac{z^3}{z^4} = z^{3-4} = z^{-1} = \frac{1}{z}$

the power of a product rule of exponents

For any nonzero real number a and natural number n, the negative rule of exponents states that

$$(ab)^n = a^n\, b^n$$

For example:

- $(ab^2)^3 = (a)^3 \cdot (b^2)^3 = a^{1\cdot 3} \cdot b^{2\cdot 3} = a^3 b^6$
- $(2t)^{15} = (2)^{15} \cdot (t)^{15} = 2^{15} t^{15}$

the power of a quotient rule of exponents

For any real numbers a and b and any integer n, the power of a quotient rule of exponents states that

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

For example:

- $\left(\frac{4}{z^{11}}\right)^3 = \frac{4^3}{(z^{11})^3} = \frac{64}{z^{11\cdot 3}} = \frac{64}{z^{33}}$
- $\left(\frac{p}{q^3}\right)^6 = \frac{p^6}{(q^3)^6} = \frac{p^{1\cdot 6}}{q^{3\cdot 6}} = \frac{p^6}{q^{18}}$

the product rule for simplifying square roots

If a and b are nonnegative, the square root of the product ab is equal to the product of the square roots of a and b.

$$\sqrt{ab} = \sqrt{a}\cdot\sqrt{b}$$

For example:

- $\sqrt{300} = \sqrt{100\cdot 3} = \sqrt{100}\cdot\sqrt{3} = 10\sqrt{3}$
- $\sqrt{162a^5b^4} = \sqrt{81a^4b^4\cdot 2a} = \sqrt{81a^4b^4}\cdot\sqrt{2a} = 9a^2b^2\sqrt{2a}$

the quotient rule for simplifying square roots

The square root of the quotient $\frac{a}{b}$ is equal to the quotient of the square roots of a and b, where $b \neq 0$.

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

For example: ● $\sqrt{\frac{5}{36}} = \frac{\sqrt{5}}{\sqrt{36}} = \frac{\sqrt{5}}{6}$

rational exponents

Rational exponents are another way to express principal nth roots. The general form for converting between a radical expression with a radical symbol and one with a rational exponent is

$$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$$

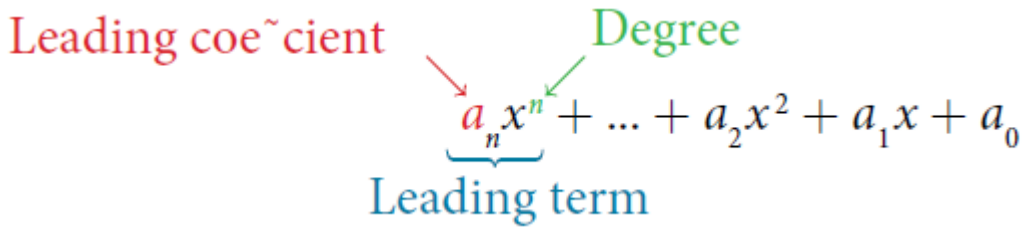
For example: ● $343^{\frac{2}{3}} = \left(\sqrt[3]{343}\right)^2 = \sqrt[3]{343^2}$

polynomials

A **polynomial** is an expression that can be written in the form

$$a_n x^n + \ldots + a_2 x^2 + a_1 x + a_0$$

Each real number a_i is called a **coefficient**. The number a_0 that is not multiplied by a variable is called a *constant*. Each product $a_i x^i$ is a **term of a polynomial**. The highest power of the variable that occurs in the polynomial is called the **degree** of a polynomial. The **leading term** is the term with the highest power, and its coefficient is called the **leading coefficient**.



perfect square trinomials

When a binomial is squared, the result is the first term squared added to double the product of both terms and the last term squared.

$$(x + a)^2 = (x + a)(x + a) = x^2 + 2ax + a^2$$

For example: ● $(3x - 8)^2 = (3x)^2 - 2(3x)(8) + (-8)^2 = 9x^2 - 48x + 64$

difference of squares

When a binomial is multiplied by a binomial with the same terms separated by the opposite sign, the result is the square of the first term minus the square of the last term.

$$(a + b)(a - b) = a^2 - b^2$$

For example:

- $(x + 5)(x - 5) = x^2 - 25$
- $(x + 11)(x - 11) = x^2 - 121$
- $(2x + 3)(2x - 3) = 4x^2 - 9$

***factoring a trinomial with leading coefficient* 1**

A trinomial of the form $x^2 + bx + c$ can be written in factored form as $(x + p)(x + q)$ where $pq = c$ and $p + q = b$.

Example: Factor $x^2 + 2x - 15$

$$x^2 + 2x - 15 = (x - 3)(x + 5)$$

Product		Sum
1	-15	-14
-1	15	14
5	-3	2
-5	3	-2

differences of squares

A difference of squares can be rewritten as two factors containing the same terms but opposite signs.

$$a^2 - b^2 = (a + b)(a - b)$$

Example: Factor $9x^2 - 25$

$$9x^2 - 25 = (3x)^2 - (5)^2 = (3x + 5)\ (3x - 5)$$

sum and difference of cubes

We can factor the sum of two cubes as

$$a^3 + b^3 = (a + b)\ (a^2 - ab + b^2)$$

We can factor the difference of two cubes as

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

For example: ● $x^3 - 2^3 = (x - 2)(x^2 + 2x + 4)$